NASA SPACE APPS CHALLENGE 2026 · BUILD A JUNIOR ASTRONAUT MISSION TRAINER

Last Light Academy

Run a Moon base. Keep the crew safe.
Help the first flower bloom on the Moon.

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THE PROBLEM

Real missions are trade-offs. Most space lessons hide them.

> Space-themed STEM content often oversimplifies the engineering trade-offs that define a real mission or presents them at a level too complex to hold a young learner's attention. Few tools make those trade-offs both tangible and fun.
>
> Challenge statement, Build a Junior Astronaut Mission Trainer (NASA Space Apps 2026)

A trade-off is only felt when it costs something you care about. So we gave the player one small, living thing to protect: Alo, a mustard plant, through one lunar night.

ONE BATTERY OF SUN-DROPS

Heat for the crew | Air maker | Grow lamp for Alo

The power that keeps the crew warm also makes their air.

ONE TANK OF WATER

Shelter wall against a solar storm | Alo's roots

The water that shields the crew from a solar storm is the water Alo drinks.

NO HELP FROM EARTH

36 hours of dark, the Earth link cut in a training exercise, and a replay of a real solar storm. The crew must decide for themselves.

THE CHALLENGE · THEME "THE NEXT FRONTIER"

Build a Junior Astronaut Mission Trainer

"Your challenge is to design and build an interactive game or app that lets students **run a lunar or Martian outpost**, balancing competing demands like **life support, radiation shielding, power, and food production**, so they **experience firsthand the decisions** that determine whether a mission fails or succeeds."

Life support

An air tank of 10 units. Four crew breathe 1 unit per 7.2-hour shift (NASA BVAD). Air at 0 stops the mission.

Radiation shielding

A water wall of up to 6 units (20 g/cm² of water) plus supply crates. Storm dose must stay at 3 or less.

Power

A battery of 20 sun-drops (≈ 10 kWh each). Daylight adds 8 per shift; the 36-hour night adds none.

Food production

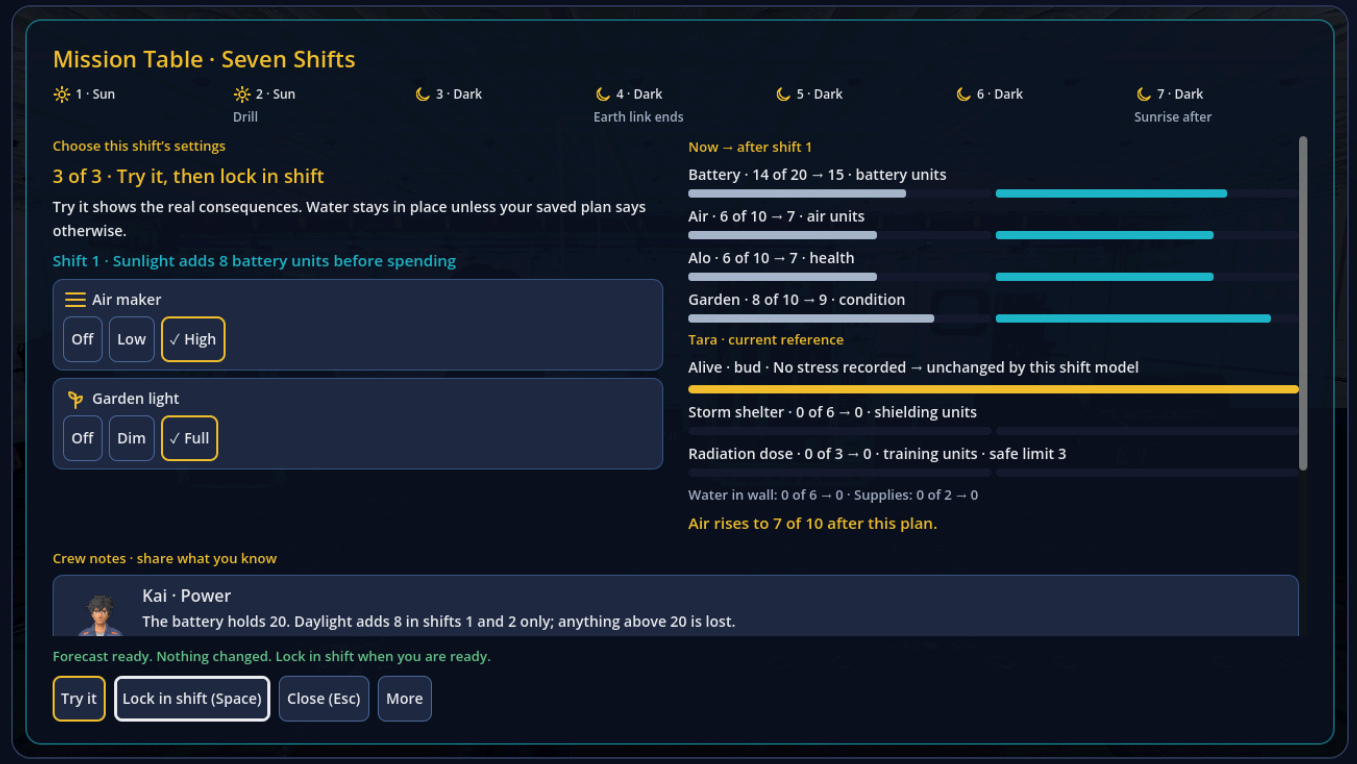
The garden, Alo's health and 12 crew-days of rations. Light and warmth cost power; water is shared with the shield.

Experience it firsthand: plan the shift → **Try it** shows the real forecast → **Lock in** → the station lives with it → four endings, a decision replay and a science debrief.

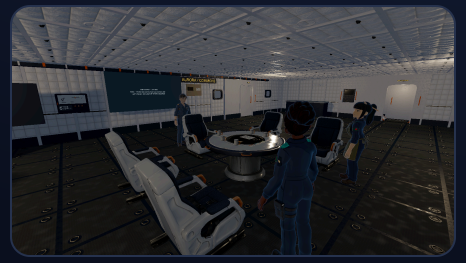
Numbers: Chapter 1 puzzle P-02 (game-design/08_PUZZLE_DESIGNS.md)

THE GAME · GODOT 4.7 · WINDOWS AND MACOS

A 3D story game where you make every call



The Mission Table (P-02). Set the air maker and garden light, press Try it, and see the real forecast: battery 14 → 15, air 6 → 7. Each cadet's note holds facts the others need.



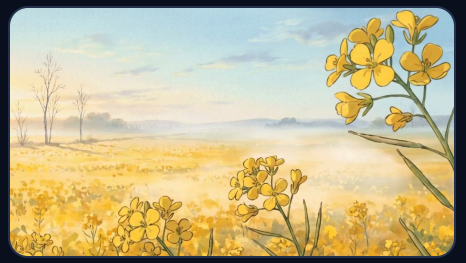
The crew in the Commons around the Mission Table



Alo, *Brassica rapa*: a real plant, never with a face



Kai, Amara, Mei, Rafi and Commander Ferreira in their EVA suits



The opening: an Earth mustard field, then a hard cut to the silent Moon

8 chapters | **32** missions | **96** scenes | **18** cutscenes | **383** voice lines, all captioned | **19** science explainers | **16** puzzles with Data Cards

All images are captures from the game (production/qa/evidence/)

NASA DATA AT THE CORE

Every key number has a source the player can open

IN THE GAME	DATASET	WHAT IT GIVES
Storm warning	NASA DONKI, 10 Sep 2017	X8.2 flare 15:35 UT; protons above 10 MeV at 16:45 UT → a **70-minute** warning
Storm curve	NOAA GOES-13 proton flux	Above 10 MeV, peak **1,493.5 pfu** on 11 Sep, 11:45 UT
Air	NASA BVAD Rev 2 (2022)	**0.895 kg** O_2 per person per day → 1 air unit = 1.07 kg
Power	NASA LaRC 10 kW south-pole array	8 sun-drops per shift = **11.1 kW**, 56 % of two arrays
Shielding	NASA LaRC (Townsend 1988; Clowdsley 2021)	**20 g/cm²** of water = the full 6-unit wall
The night	LRO LOLA illumination (Mazarico 2011)	Longest darkness on Shackleton's rim ≈ 1.5 days → a **36-hour** night
Ground and sky	LRO LOLA elevation; JPL Horizons	Aurora at **89.44° S**, 1,959.5 m; Earth below the pole's horizon 15–28 Apr 2046
Storm at the Moon	LRO CRaTER dose, Jul 2012	0.57 → **3.92 mGy/day** (×6.8) in Chapter 7's "what if"



An in-game explainer, drawn by code from the fixture numbers: GOES-13 protons and the DONKI flare time.

OBSERVED · COMPUTED · TRAINING ASSUMPTION

29 sources ship offline: 21 NASA and 8 partner (NOAA, USGS/IAU, ESA, ISRO, JAXA, CNSA, UW–Madison). Each Data Card names the dataset, what was measured, what the game computed, and the link. Numbers no dataset supports are tagged as gameplay values and listed publicly.

Every row is a JSON fixture in game/resources/data/nasa/ · full map: docs/SCIENCE_AND_DATA.md §B

LEARNING EVIDENCE

Built into every night. Measured with learners next.



After the night: predict → what happened → why, then a check question. A replay shows each shift's choice, its forecast and what really happened.

Learner playtest PENDING · BOARD M1-22

- 5 or more first-time players in the target age range, minors only with a guardian present
- Time to the first objective
- P-02 attempts and the ending reached
- Can they explain the numbers? One predict and one explain question each
- Where they got confused

Results: [metric from M1-22 when the playtest runs]

What we can show today

- 58 automated test suites; the latest full run passed 57 (one loading-time gate is being fixed)
- About 60 fps on an Apple M4 Mac in the timed rooms; other hardware not measured yet
- Captions on and nothing timed by default; 1.5× text; reduced motion

Debrief and decision replay: built and reviewed; final test gate pending (board M1-19, M1-68)

IMPACT AND NEXT STEPS

A free mission trainer any classroom can run

Who it reaches

- Learners aged 10 to 15, in classrooms and at home
- Runs offline on Windows and macOS, no account, no internet
- Free and open source (Apache-2.0): schools can copy, translate and adapt it
- One Chapter 1 night is designed to take about 20 minutes: it fits in a lesson
- From Bangladesh to the Moon: Alo is *Brassica rapa*, the species of Bangladesh's mustard and of the Fast Plants grown in classrooms worldwide

Next steps

- Four-player Class mode: shared decisions, private role clues on one machine (in progress, M1-37)
- Interactive puzzle panels for Chapters 2–8 (M3-2P to M3-8P)
- The learner playtest, then fixes from what players show us (M1-22, M1-24)
- The release build for judging on 12 November (M3-93)
- Two registered sources still to process: the LROC illumination map and the 2017 CRaTER storm curve

Code, builds and data: github.com/SadManQT/nasa-space-apps-2026 · **Team:** spaceappschallenge.org/2026/find-a-team/the-last-braincell

Made with AI tools, every generation logged in production/AI_PROMPT_LOG.md: Claude Code (Anthropic), OpenAI Codex, Google Antigravity, Nano Banana Pro (Gemini 3 Pro Image), GPT Image 2.5, Topaz Image Upscale, Tripo AI, Meshy, ElevenLabs, Seedance 2.5, Google Veo 3.1. The story, dialogue, puzzles and numbers, science mapping and art direction are the team's.

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